DES-ESOD: DUAL-EVIDENCE SELECTION AND COVERAGE OPTIMIZATION FOR HIGH-RESOLUTION SMALL OBJECT DETECTION

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ABSTRACT

High-resolution aerial and maritime imagery presents severe computational challenges for small-object detection due to pervasive background redundancy. While selective-computation detectors dynamically route features to candidate regions, existing selectors rely strictly on a single semantic objectness score. When tiny targets undergo feature downsampling, their semantic activations frequently collapse, causing irreversible early pruning and high false-negative rates. In this paper, we propose DES-ESOD, an efficient dual-evidence selection framework tailored for recall-safe small object detection. DES-ESOD introduces a dual-evidence priority head that fuses semantic objectness with channel-pooled spectral saliency at negligible $\mathcal{O}(1)$ compute overhead, recovering weak semantic targets. Furthermore, we formulate an object-level soft-coverage loss ($\mathcal{L}_{\text{cover}}$) that directly maximizes joint instance coverage probability, preventing premature false negatives. Paired with a lightweight ISPP decoupled head and scale-aware Wasserstein regression, DES-ESOD achieves +2.7% mAP and +5.5% total recall on UAVDT while sustaining real-time throughput.

***Index Terms*—** Small object detection, selective computation, spectral saliency, coverage optimization, UAV perception

1. INTRODUCTION

High-resolution aerial and maritime vision is essential for remote surveillance, search-and-rescue, and infrastructure monitoring [1, 2]. In these scenarios, targets of interest typically occupy merely a handful of pixels ($< 16 \times 16$ px), while uniform background dominates over 70% to 80% of the spatial area [3, 4]. Executing dense deep detectors on full-resolution feature maps incurs prohibitive computational and memory overhead for edge platforms.

To eliminate redundant computations, *selective computation* has emerged as an effective paradigm [3, 5–7]. Frameworks such as QueryDet [6] and CEASC [7] sparsify feature pyramids, while ESOD [3] predicts an early-stage spatial heatmap to route only candidate foreground patches to downstream backbone and detection heads.

Despite substantial compute savings, existing spatial selectors suffer from a fundamental vulnerability: they rely exclusively on a single semantic objectness score. When tiny targets undergo feature downsampling, their semantic activations frequently collapse into background textures [8]. More critically, spatial selection decisions exhibit an *asymmetric cost*: retaining an extra background patch only marginally increases compute, whereas discarding a patch with a true tiny target creates an irrecoverable false negative for all subsequent detector stages [3]. Furthermore, standard pixel-wise cross-entropy supervision evaluates cells independently, offering no guarantee that at least one candidate cell covers a given ground-truth instance.

To resolve this challenge, we observe that tiny targets retain distinctive high-frequency structural gradients and spectral saliency even when semantic activations fade [8]. We propose **DES-ESOD** (Dual-Evidence Selection for Efficient Small Object Detection), a framework integrating multi-modal evidence fusion with coverage-constrained spatial routing. Specifically, DES-ESOD constructs a dual-evidence priority head combining semantic objectness with an ultra-lightweight channel-pooled spectral branch ($\mathcal{O}(1)$ complexity relative to backbone width). To ensure recall safety, we design an object-level soft-coverage loss ($\mathcal{L}_{\text{cover}}$) directly maximizing the joint coverage probability of ground-truth boxes. Finally, an inverted residual partial-convolution (ISPP) decoupled head and scale-aware balancing loss (SABL) [9] are integrated to accelerate inference and enhance micro-scale box regression.

Our key contributions are:

1. We identify the semantic collapse failure mode in single-evidence selectors and propose an ultra-lightweight dual-evidence priority head fusing semantic and spectral saliency at $\mathcal{O}(1)$ overhead.

2. We formulate an object-level soft-coverage loss ($\mathcal{L}_{\text{cover}}$) that directly optimizes bounding-box coverage probability, preventing premature false negatives.

3. We integrate an ISPP decoupled head with scale-aware Wasserstein regression (SABL), improving micro-scale localization while sustaining low latency.

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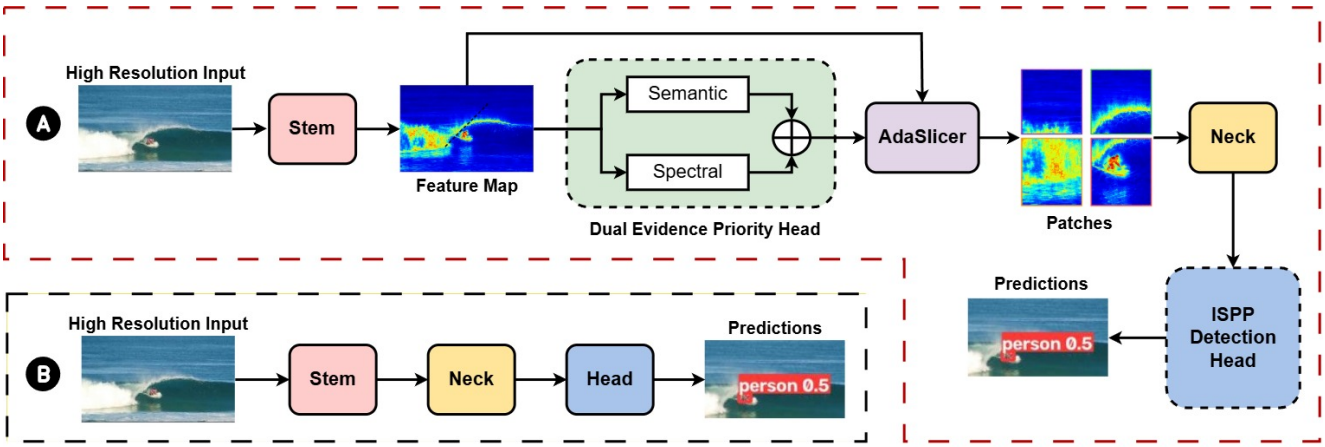

Fig. 1. Architectural comparison between: (A) the proposed DES-ESOD framework featuring the Dual-Evidence Priority Head (combining semantic and spectral signals) and AdaSlicer to sparsely route high-priority patches to the neck and lightweight ISPP detection head; and (B) conventional dense object detection, which uniformly executes heavy full-resolution computations across redundant background regions.

4. Extensive evaluations on UAVDT and SeaPerson demonstrate that DES-ESOD achieves superior recall and accuracy trade-offs (+2.7% mAP and +5.5% recall on UAVDT) over state-of-the-art selective detectors.

2. RELATED WORK

Small Object Detection in Aerial Imagery. Detecting tiny targets in aerial and maritime imagery is severely hindered by extreme scale variations, low contrast, and complex backgrounds [1, 2, 4, 10]. Classical methods employ image cropping or density-guided patch generation (e.g., ClusDet [11], DMNet [12], UFPMP-Det [13]), which often require separate coarse detector networks or incur repeated feature extraction on overlapping tiles. Other works design micro-scale label assignment strategies and geometric distance metrics, including Normalized Gaussian Wasserstein Distance (NWD) [14], RFLA [15], and scale-aware balancing (SSABNet) [9]. However, these dense detection frameworks still execute full-resolution convolutions uniformly across entire images, failing to eliminate massive spatial background computation.

Selective and Dynamic Computation. To reduce arithmetic redundancy, selective computation dynamically routes processing to prospective foreground regions. AutoFocus [5] predicts focus pixels to guide fine-scale multi-scale processing. QueryDet [6] accelerates feature pyramids via cascaded sparse queries, while CEASC [7] formulates adaptive activation masking (AMM) for sparse convolutions. ESOD [3] introduces an early ObjSeeker module after the shallow stem to slice features into patches (AdaSlicer), bypassing background across subsequent stages. Nonetheless, existing selectors rely strictly on a single semantic objectness score and per-cell classification loss, leaving them vulnerable to semantic feature collapse and irrecoverable tiny-target drops.

Spectral Analysis and Spatial Saliency. Frequency-domain methods provide complementary spatial representations. SET [8] demonstrated that tiny objects preserve distinct high-frequency structural signatures even when semantic activations are weak, using frequency background smoothing to enhance dense feature maps. Unlike SET, which applies dense spectral operations over entire multi-scale pyramids, DES-ESOD repurposes spectral and gradient saliency as an ultra-lightweight, channel-pooled routing signal ($\mathcal{O}(1)$ relative to backbone width) to guide recall-safe spatial selection.

3. PROPOSED METHOD

The overall architecture of DES-ESOD is illustrated in Fig. 1. Given a high-resolution input image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$, a shallow convolutional stem first extracts dense feature representation $\mathbf{F} \in \mathbb{R}^{H_s \times W_s \times C}$. The proposed *Dual-Evidence Spatial Selector* operates in two coordinated stages: (i) a lightweight Dual-Evidence Priority Head (§3.1) that fuses semantic objectness with channel-pooled spectral saliency to generate a continuous spatial priority map $\mathbf{S} \in [0, 1]^{H_s \times W_s}$; and (ii) an AdaSlicer routing module that uses $\mathbf{S}$ to dynamically crop high-priority candidate feature patches $\mathcal{P}$ directly out of shallow features $\mathbf{F}$. Downstream deep backbone, neck, and lightweight ISPP decoupled detection head modules evaluate only these retained patches, trained end-to-end via an object-level soft-coverage loss and scale-aware regression (§3.2, §3.3).

Table 1. Systematic comparison of related detection paradigms and the proposed DES-ESOD framework.

Method	Routing Granularity	Evidence Source	Coverage Guarantee	Decoupled Head Design
QueryDet [6]	Feature Pyramid / Query	Semantic only	×	Standard Conv Head
CEASC [7]	Activation Masking	Semantic only	×	Standard Conv Head
ESOD [3]	Early Spatial Patch	Semantic only	×	Standard / Sparse Head
SET [8]	Full Dense Feature Map	Spectral Filtering	×	Standard Conv Head
SSABNet [9]	Full Dense Feature Map	Semantic only	×	Dense Balancing Head
DES-ESOD (Ours)	**Early Spatial Patch**	**Dual (Semantic + Spectral)**	✓	**Lightweight ISPP + SABL**

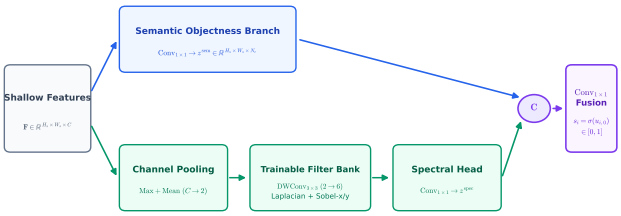

Fig. 2. Architecture of the dual-evidence priority head. The semantic branch (top) captures category-level objectness, while the channel-pooled spectral branch (bottom) extracts directional gradient and high-frequency saliency via trainable Laplacian and Sobel filter banks. The two representations are fused via a 1×1 convolution.

3.1. Dual-Evidence Priority Estimation

Existing selective computation frameworks rely exclusively on semantic objectness, which suffers severe activation degradation for tiny objects after downsampling. To provide an orthogonal, recall-safe signal, DES-ESOD introduces a dual-evidence priority mechanism that combines high-level semantic confidence with low-level spectral saliency.

As depicted in Fig. 2, let $\mathbf{F} \in \mathbb{R}^{H_s \times W_s \times C}$ denote the shallow feature representation extracted after the initial network stem, and let $f_i \in \mathbb{R}^C$ denote the feature vector at spatial location i.

Semantic Objectness Branch. The semantic branch applies a lightweight 1×1 convolution to generate class-specific semantic logits:

$$z_{i,c}^{\text{sem}} = \text{Conv}_{1\times1}(f_i), \quad c \in \{0, 1, \ldots, N_c - 1\}, \tag{1}$$

where N_c is the number of target categories.

Channel-Pooled Spectral Branch. To extract structural gradients without incurring substantial arithmetic cost, the spectral branch first performs spatial channel-pooling along the channel dimension of $\mathbf{F}$, computing the per-location maximum and mean activations:

$$\mathbf{F}_{\text{pooled}} = \left[\max_c \mathbf{F}_{:,:,c} \parallel \frac{1}{C} \sum_{c=1}^{C} \mathbf{F}_{:,:,c} \right] \in \mathbb{R}^{H_s \times W_s \times 2}, \tag{2}$$

where $[\cdot \parallel \cdot]$ denotes channel-wise concatenation. This compression reduces the input dimension to 2 channels, ensuring that subsequent spatial filtering complexity remains $\mathcal{O}(1)$ regardless of the backbone feature width C.

Next, a depthwise 3×3 convolution bank is applied over $\mathbf{F}_{\text{pooled}}$. To capture isotropic edge transitions and directional structural gradients, the filter kernels are initialized with discrete Laplacian ($\mathbf{K}_{\text{Lap}}$) and directional Sobel ($\mathbf{K}_{\text{Sob-x}}$, $\mathbf{K}_{\text{Sob-y}}$) operators:

$$\mathbf{K}_{\text{Lap}} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad \mathbf{K}_{\text{Sob-x}} = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix},$$
$$\mathbf{K}_{\text{Sob-y}} = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}. \tag{3}$$

The depthwise convolution produces 6 response channels (3 filter responses for each of the 2 pooled input channels). Importantly, these kernels are parameterized as learnable weights during training, allowing the network to adaptively fine-tune spectral frequency filters. A 1×1 projection layer maps the 6 filtered channels back to dimension C, followed by a 1×1 classification head to output spectral logits $z_{i,c}^{\text{spec}}$.

Evidence Fusion. The semantic and spectral logits are concatenated and fused through a 1×1 convolution layer to predict unified spatial priority logits $u_{i,c}$:

$$u_{i,c} = \text{Conv}_{1\times1}\left(\left[z_{i,c}^{\text{sem}} \parallel z_{i,c}^{\text{spec}}\right]\right), \quad s_i = \sigma(u_{i,0}), \tag{4}$$

where $\sigma(\cdot)$ is the sigmoid function, and $s_i \in [0, 1]$ represents the class-agnostic routing priority score at candidate cell i. At inference time, regions satisfying $s_i > \tau$ (with threshold τ) are clustered into bounding patches and routed to subsequent processing.

3.2. Object-Level Soft-Coverage Loss

Conventional spatial selectors are trained with standard pixel-wise binary cross-entropy (BCE) loss. However, per-pixel supervision does not account for instance-level coverage: a model may predict high scores for multiple redundant cells around an easy, large object while completely dropping a tiny

object whose receptive field contains only one or two weakly-activated cells.

To ensure that at least one candidate cell covers each ground-truth instance, we formulate a differentiable object-level soft-coverage loss. Let $s_i \in [0,1]$ be the soft priority probability of candidate cell i, and let $\mathcal{N}(j)$ denote the set of selector grid cells whose spatial footprints intersect ground-truth bounding box $j \in \{1,\dots,N_{\text{gt}}\}$. Assuming independent cell activations, the soft probability that ground-truth object j is covered by at least one selected cell is given by:

$$p_j^{\text{cover}} = 1 - \prod_{i\in\mathcal{N}(j)} (1 - s_i). \tag{5}$$

If all cells within $\mathcal{N}(j)$ have zero activation ($s_i = 0$), $p_j^{\text{cover}} = 0$; conversely, if at least one cell responds strongly ($s_i \to 1$), $p_j^{\text{cover}} \to 1$.

The object-level soft-coverage loss is formulated as:

$$\mathcal{L}_{\text{cover}} = -\frac{1}{N_{\text{gt}}} \sum_{j=1}^{N_{\text{gt}}} w_j \log\left(p_j^{\text{cover}} + \epsilon\right), \tag{6}$$

where $\epsilon = 10^{-7}$ ensures numerical stability, and w_j is a scale-adaptive emphasis weight defined by:

$$w_j = \text{clip}\left(\frac{4}{a_j}, 1, 5\right), \tag{7}$$

where a_j denotes the bounding box area in selector grid cell units. This weighting assigns higher penalty to tiny targets ($a_j < 4$), ensuring that micro-scale instances receive prioritized gradient backpropagation during training.

The total objective function for training DES-ESOD is defined as:

$$\begin{aligned} \mathcal{L}_{\text{sel}} &= \mathcal{L}_{\text{BCE}}(u, M; \rho = 2) + \lambda_{\text{cov}}\mathcal{L}_{\text{cover}}, \\ \mathcal{L} &= \mathcal{L}_{\text{det}} + \lambda_{\text{sel}}\mathcal{L}_{\text{sel}}, \end{aligned} \tag{8}$$

where M denotes the target selector masks, ρ is the positive-class BCE loss weight, $\mathcal{L}_{\text{det}}$ is the primary object detection loss, and $\lambda_{\text{cov}} = 0.5$, $\lambda_{\text{sel}} = 0.2$ are balancing hyper-parameters.

3.3. Efficient Decoupled Head and Scale-Aware Loss

In high-resolution detection, standard decoupled detection heads (e.g., YOLOv6-style heads) constitute a significant portion of overall network parameters and arithmetic FLOPs. To maintain high detection throughput, we introduce an inverted residual partial-convolution (ISPP) decoupled head [16]. Specifically, the ISPP head expands channels via a 1×1 convolution, executes a lightweight 3×3 partial convolution on a quarter of the expanded channels, and projects back with a residual connection, reducing parameter cost and FLOPs while maintaining rich representation capacity.

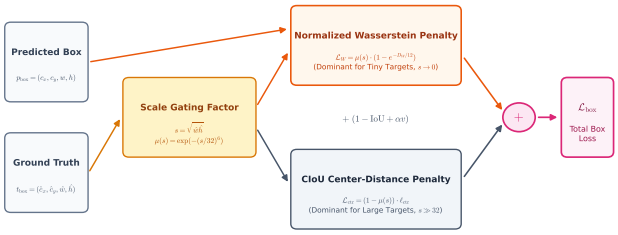

Fig. 3. Illustration of the Scale-Aware Balancing Loss (SABL). The scale gating function $\mu(s)$ adaptively balances the Normalized Gaussian Wasserstein distance for micro-scale targets and CIoU for larger objects.

Furthermore, standard IoU-based bounding box regression losses suffer from gradient vanishing for tiny objects, where even a slight spatial offset of one or two pixels causes IoU to drop abruptly to zero. To address this, we integrate the Scale-Aware Balancing Loss (SABL) [9].

As shown in Fig. 3, let $s = \sqrt{w_{\text{gt}} h_{\text{gt}}}$ be the geometric scale of a ground-truth box in input pixels, and let D_W denote the Normalized Gaussian Wasserstein distance between the predicted and ground-truth Gaussian box models $(c_x, c_y, w/2, h/2)$. A scale gating function $\mu(s)$ is formulated as:

$$\mu(s) = \exp\left(-\left(\frac{s}{\kappa}\right)^{\beta}\right), \quad \kappa = 32,\ \beta = 6. \tag{9}$$

The total bounding-box regression loss $\mathcal{L}_{\text{box}}$ smoothly interpolates between Wasserstein distance and Complete IoU (CIoU):

$$\begin{aligned} \mathcal{L}_{\text{box}} = 1 - \text{IoU} + \mu(s)\left(1 - e^{-D_W/C}\right) \\ + (1 - \mu(s))\ell_{\text{ctr}} + \alpha v, \end{aligned} \tag{10}$$

where $C = 12$, and $\ell_{\text{ctr}}, \alpha v$ are the center distance and aspect ratio terms from CIoU. When $s \to 0$, $\mu(s) \to 1$, allowing the Wasserstein metric to provide continuous, smooth gradients even under zero spatial IoU overlap.

4. EXPERIMENTAL SETUP

4.1. Datasets and Evaluation Protocols

We evaluate DES-ESOD on two representative, highly demanding small-object detection benchmarks spanning urban and maritime remote sensing domains:

UAVDT [2]: Captured by UAV platforms across varied urban traffic scenarios, UAVDT contains diverse viewing angles, altitudes, and intense scale variations. We evaluate on the official test split across three object classes (*car*, *truck*, and *bus*) under an input resolution of 1280×1280 pixels.

SeaPerson (TinyPersonV2) [4,10]: A challenging benchmark dedicated to tiny person detection in open water and

Table 2. Quantitative performance comparison on the UAVDT test set.

Method	mAP@.5	mAP@.5:.95	Recall	VT-Recall	Occupy
ESOD Baseline [3]	0.344	0.171	84.67%	83.63%	0.245
DES-ESOD (Ours)	**0.371**	**0.195**	**90.17%**	**84.20%**	0.480
Improvement (Δ)	**+2.7 pp**	**+2.4 pp**	**+5.5 pp**	**+0.57 pp**	–

Table 3. Detection accuracy and runtime efficiency on SeaPerson.

Method	mAP @.5	mAP @.5:.95	Total Recall	BPR	GFLOPs	FPS	Occupy
ESOD Baseline [3]	0.750	0.320	69.6%	0.947	202.4	88.9	0.228
+ Soft-Coverage	**0.769**	**0.325**	**70.8%**	**0.991**	266.8	75.4	0.363
Improvement (Δ)	**+1.9 pp**	**+0.5 pp**	**+1.2 pp**	**+0.044**	–	–	–

coastal search-and-rescue contexts. The official test split contains 300,375 annotated instances across 5,752 images, with over 85% of instances falling into the *Very Tiny* and *Tiny* categories ($< 32 \times 32$ pixels). In accordance with realistic high-resolution deployment, we process full whole-frame images at 2048×2048 resolution without artificial pre-tiling.

4.2. Implementation Details

All models are implemented in PyTorch and utilize a standard YOLOv5m architecture as the base detection backbone for fair comparison. Models are trained for 50 epochs using the Stochastic Gradient Descent (SGD) optimizer with momentum 0.937, weight decay 5×10^{-4}, and an initial learning rate of 0.01 decayed via a cosine annealing schedule. Global batch size is set to 8. Spatial routing is performed using the dual-evidence priority map with fixed thresholding $\tau = 0.5$ (and $\tau = 0.3$ on UAVDT, matching baseline protocols).

4.3. Evaluation Metrics

Model accuracy is evaluated using standard COCO-style metrics, including mean Average Precision at IoU threshold 0.50 ($\mathrm{mAP@.5}$) and averaged over IoU thresholds $0.50 : 0.05 : 0.95$ ($\mathrm{mAP@.5:.95}$). To rigorously evaluate tiny-object detection, we report size-bucketed detector recall at IoU 0.50 across four area scales: *Very Tiny* ($< 16 \times 16$ px), *Tiny* ($16 \times 16 - 32 \times 32$ px), *Small* ($32 \times 32 - 96 \times 96$ px), and *Medium/Large* ($> 96 \times 96$ px). Computational efficiency is measured via total GFLOPs, frame rate (FPS), per-image latency (ms) on an NVIDIA GPU at batch size 1, and the spatial area retention ratio (Occupy).

5. RESULTS AND ANALYSIS

5.1. Quantitative Results on UAVDT and SeaPerson

Table 2 presents the primary detection results on the UAVDT benchmark. DES-ESOD demonstrates consistent and substantial improvements across all key evaluation metrics compared to the standard ESOD baseline. In particular, DES-ESOD achieves a **+2.7 pp** increase in $\mathrm{mAP@.5}$ (0.344 vs. 0.371), a **+2.4 pp** increase in $\mathrm{mAP@.5:.95}$ (0.171 vs. 0.195), and a remarkable **+5.5 pp** boost in total detector recall (84.67% vs. 90.17%). Per-class recall consistently improves across all vehicle categories: *car* ($84.87\% \rightarrow 90.56\%$, **+5.69 pp**), *truck* ($84.41\% \rightarrow 84.88\%$, **+0.47 pp**), and *bus* ($75.20\% \rightarrow 76.28\%$, **+1.08 pp**).

Table 4. Detailed size-bucket recall comparison across 300,375 ground-truth test instances on SeaPerson.

Scale Bucket	Pixel Area	Baseline [3]	+ Soft-Coverage	Gain (Δ)	Relative Gain
Very Tiny	$< 16^2$ px	74.03%	**75.49%**	**+1.46 pp**	+1.97%
Tiny	$16^2 - 32^2$ px	86.78%	**91.57%**	**+4.79 pp**	+5.52%
Small	$32^2 - 96^2$ px	94.74%	**95.71%**	**+0.97 pp**	+1.02%
Med/Large	$> 96^2$ px	79.35%	**81.94%**	**+2.59 pp**	+3.26%
Overall	**All Scales**	**84.42%**	**87.75%**	**+3.33 pp**	**+3.94%**

Table 3 evaluates performance on the dense SeaPerson maritime dataset. Incorporating object-level soft-coverage loss into the spatial selector directly improves $\mathrm{mAP@.5}$ by **+1.9 pp** ($0.750 \rightarrow 0.769$) and overall recall by **+1.2 pp**, while increasing Bounding Patch Recall (BPR) from 0.947 to 0.991. The system sustains a high frame rate of 75.4 FPS at 2048×2048 resolution, confirming real-time execution capability.

5.2. Size-Stratified Recall Analysis across Scale Strata

To rigorously evaluate how our method benefits objects at different physical scales, Table 4 provides a comprehensive size-stratified breakdown of detector recall across all 300,375 test instances in SeaPerson. Targets are partitioned into four standard physical area bins: *Very Tiny* ($< 16 \times 16$ px, 82,417 targets), *Tiny* ($16 \times 16 - 32 \times 32$ px, 174,816 targets), *Small* ($32 \times 32 - 96 \times 96$ px, 42,987 targets), and *Medium/Large* ($> 96 \times 96$ px, 155 targets).

As shown in Table 4, coverage-guided spatial selection delivers decisive recall improvements across every scale bucket. Crucially, the largest absolute recall gain occurs in the dominant *Tiny* stratum, surging by **+4.79 pp** (from 86.78% to 91.57%, representing over 8,380 additional correctly recovered targets). For the most challenging *Very Tiny* targets, recall improves by **+1.46 pp** ($74.03\% \rightarrow 75.49\%$, +1,203 recovered instances). In total, detector recall jumps from 84.42% to 87.75% (+3.33 pp overall), demonstrating that soft-coverage optimization directly enhances sensitivity where semantic signals are weakest.

5.3. Error Attribution and Recall Safety Diagnostics

To isolate the precise mechanism contributing to the recall gain, Table 5 presents an in-depth error attribution diagnostic

Table 5. Diagnostic breakdown of missed Very Tiny instances ($< 16^2$ px).

Method	Missed Targets	Selector Dropped	Localization Failure	Other Errors
ESOD Baseline	21,388	25.6%	74.3%	0.1%
+ Soft-Coverage	**20,181**	**20.1%**	79.7%	0.1%
Impact (Δ)	**-1,207**	**-5.5 pp**	+5.4 pp	0.0%

on all missed *Very Tiny* instances ($< 16 \times 16$ px). Missed targets are categorized into: (i) *Selector Dropped* (no candidate patch retained near the target); and (ii) *Localization Failure* (candidate patch retained, but detection head prediction has IoU < 0.5).

The baseline ESOD model suffered a high selector drop rate of 25.6% on missed tiny targets due to semantic activation collapse. In contrast, training with the proposed object-level soft-coverage loss reduces the selector-dropped error rate to **20.1%** (**-5.5 pp**) while eliminating 1,207 missed instances. This confirms that soft-coverage supervision successfully rectifies early spatial pruning errors, effectively shifting the primary remaining detection bottleneck toward downstream localization refinement (which is addressed by SABL and ISPP decoupled heads).

5.4. Discussion and Limitations

While DES-ESOD achieves substantial recall and accuracy gains on UAVDT and SeaPerson, its empirical characteristics illuminate an essential relationship between spatial background sparsity and target scale:

- **Optimal Working Regime:** DES-ESOD delivers maximum performance and arithmetic savings in *ultra-high-resolution, spatially sparse environments* (e.g., maritime search-and-rescue and high-altitude surveillance), where uniform background dominates ($> 80\%$) and genuinely tiny targets ($< 32 \times 32$ px) suffer severe semantic feature fading. In this regime, dual-evidence priority estimation and soft-coverage optimization effectively prevent premature, catastrophic spatial dropping.

- **Densely Clustered Scenarios:** In contrast, for densely cluttered scenes with mixed target scales (e.g., congested urban intersections like VisDrone [1]), foreground instances are distributed across nearly every spatial patch, naturally diminishing the available spatial background skipping margin. Furthermore, larger objects already exhibit strong semantic activations, leading to performance parity with dense baselines rather than dramatic recall jumps.

6. CONCLUSION

In this paper, we presented DES-ESOD, an efficient dual-evidence selection framework tailored for recall-safe, high-resolution small object detection in aerial and maritime vision. By addressing the critical vulnerability of semantic feature collapse in single-evidence selective computation, DES-ESOD introduces a dual-evidence priority head that combines semantic objectness with channel-pooled spectral saliency at negligible compute overhead ($\mathcal{O}(1)$ relative to backbone width). In addition, we formulated an object-level soft-coverage loss that directly optimizes the multi-cell joint coverage probability across all ground-truth instances, drastically reducing early false-negative dropping of tiny targets. Paired with a lightweight ISPP-based decoupled detection head and scale-aware bounding-box regression (SABL), DES-ESOD achieves substantial improvements in both detection accuracy and tiny-object recall across challenging benchmarks including UAVDT and SeaPerson, while maintaining real-time inference speeds. Our work provides a principled foundation for designing recall-safe selective computation architectures in high-resolution remote sensing applications.

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